

JASH SAHA

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ABOUT ME

I am a responsible Engineering student with strong problem-solving and project management skills. I focus on solving practical problems. My background combines technical rigor with a structured approach to project management, ensuring efficient and business-aligned delivery.

EDUCATION

❖ DELHI PUBLIC SCHOOL,RUBY PARK

High School Diploma in Science

(Feb 2009 – Feb 2023)

❖ MEGHNAD SAHA INSTITUTE OF TECHNOLOGY KOLKATA

Bachelor of Technology in Computer Science and Business Systems

(Aug 2023 - July 2027)

SKILLS

❖ Technical Skills

- C, C++, Python, SQL, HTML & CSS, React.js, Node.js & JavaScript.
- PyTorch, Tensorflow, Scikit-Learn, OpenCV, MediaPipe, Pandas, NumPy & Matplotlib.
- Data cleaning, Data engineering, Data Visualization & analysis.

❖ Other Skills

- Public Speaking, Leadership, Communication, Team work & Management.

PROJECTS, RESEARCH & PATENTS

❖ Sign language translation system for speech impaired people (**Published Patent on 27th March 2026**)

- **Bridging the Gap:** To facilitate seamless communication between sign language users and non-signers by eliminating language barriers.
- **Real-Time Translation:** To instantly detect hand gestures using computer vision and convert them into spoken audio.

❖ ML-based prediction model for sensitivity of a gate overlapped heterojunction TFET biosensor (**Research in progress**)

- **Ensemble Machine Learning:** Implemented a Random Forest Regressor to model complex non-linear dependencies in multi-variable datasets, optimizing feature selection for accurate predictive modeling.
- **Automated Data Filtering:** To automatically detect and remove statistical outliers by validating experimental data.

❖ End-to-end AI system for predicting agricultural commodity prices and assessing market risks

- **Designed and trained a custom PyTorch Neural Network:** Utilizing a Multi-Algorithm Ensemble (MoE) architecture to predict localized agricultural commodity prices
- **Developed a dynamic risk-assessment algorithm:** That translates AI predictions into actionable market intelligence, categorizing supply chain risks (e.g., Critical Shortage, Elevated Risk) and forecasting price trends.

❖ Bad Touch Detection System

- **Real-Time Computer Vision Pipeline:** Engineered a safety monitoring system using OpenCV and MediaPipe that tracks skeletal landmarks to differentiate between "self-touch" and potentially unsafe "foreign contact" with high precision.
- **Automated Threat Response:** Implemented a geometric logic algorithm to define safety zones around the torso, triggering an immediate audio alarm via Pygame if unauthorized proximity is detected, ensuring proactive personal safety.

❖ Secure Full-Stack Real-Time Chat Application

- **Scalable Backend & Database Integration:** Engineered an asynchronous Python server (aiohttp, Socket.IO) integrated with PostgreSQL to seamlessly handle concurrent connections, room-based message routing, and persistent chat history.
- **Identity Protection & Authentication:** Implemented dynamic user registration and authentication utilizing bcrypt password hashing to secure private chat rooms and protect user identities.
- **WebRTC & Multimedia Capabilities:** Designed a responsive, full-screen Progressive Web App (PWA) featuring WebRTC for low-latency peer-to-peer audio/video calling and real-time file sharing.

PROFESSIONAL EXPERIENCE

❖ Indian Institute of Internship (***Business Analytics Intern***)

Built Power BI dashboards to analyze sales data & improve reporting efficiency.

(Aug 2025 – Oct 2025)

ADDITIONAL INFORMATION (***Leadership & Management***)

❖ Activities:

- **President & Ex-Vice President of Phoenix(Cultural committee of CSBS):-** (Nov 2024 – Current)
- **Vice President & Ex-Speaker of Greenovation Club(GCMSIT):-** (April 2025 – Current)